ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET Mechatronic Systems paper (project based, 100 marks): design constraints and specifications, a hydraulic baling press, cylinder force and speed calculations, hydraulic circuits, interlocks, bills of materials, Gantt charts, PID control, system block diagrams, commissioning a hydraulic system, maintenance schedules and hydraulic safety.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers.
5. Draw sketches, charts and circuits neatly. Use only a black or blue pen.

> **PROJECT.** A recycling centre packs loose cardboard into bags by hand. The manager wants a hydraulic baling press: cardboard is loaded through a door at the front, a double-acting hydraulic cylinder pushes a platen down to compress it, and a pressure switch ends the stroke when the bale is tight. The press may only operate when the door is closed. A PLC controls the press and a hydraulic power pack supplies the oil.

QUESTION 1

1.1 Explain what a constraint is in a design project, and give THREE constraints for the baling press. (4)

1.2 Write SIX specifications for the baling press. (6)

[10]

QUESTION 2

2.1 Make a neat freehand sketch of the baling press. Show the frame and baling chamber, the loading door with its interlock switch, the hydraulic cylinder and platen, the power pack and the control panel. (5)

2.2 The cylinder has a bore of 100 mm and a rod diameter of 50 mm. The system pressure is 12 MPa and the pump delivers 20 litres per minute. Calculate the:

2.2.1 compression force of the cylinder (3)

2.2.2 extending speed of the cylinder in mm/s (3)

2.2.3 retracting speed of the cylinder in mm/s (2)

2.3 Draw the hydraulic circuit of the baling press. Show the tank, filter, pump, relief valve and pressure gauge, a 4/3-way double-solenoid valve with a tandem centre and spring centring, and the double-acting cylinder. (8)

2.4 Explain why a pressure switch, rather than a limit switch, is used to end the compression stroke. (2)

2.5 Name TWO safety interlocks that the baling press must have. (2)

[25]

QUESTION 3

3.1 Draw up a bill of materials for the baling press with suitable column headings and at least SIX items. (8)

3.2 Draw a Gantt chart for SIX activities of the project over a period of 6 weeks. Some activities must overlap. (10)

3.3 The manager wants the pressure to be held constant with a proportional valve and a PID controller. Explain the effect of the P, the I and the D part of the controller. (6)

3.4 Draw a system block diagram of the baling press, showing its inputs, controller and outputs. (6)

[30]

QUESTION 4

4.1 Explain why the relief valve must be set before the press is used for the first time. (2)

4.2 Use a spider diagram to show SIX areas that must be tested on the baling press. (6)

4.3 Describe how the setting of the pressure switch is adjusted during commissioning. (3)

4.4 Explain the purpose of a dry run (without cardboard) before production starts. (2)

4.5 Name TWO items of personal protective equipment that must be worn while a hydraulic system is commissioned. (2)

[15]

QUESTION 5

5.1 Draw up a preventive maintenance schedule for the baling press with FOUR tasks and how often each must be done. (8)

5.2 Give THREE possible causes of hydraulic oil overheating. (3)

5.3 Describe the steps to make the press safe before maintenance is done on the hydraulic system. (4)

5.4 Explain why a leaking high-pressure hose must never be checked with a bare hand. (2)

5.5 Explain how oil analysis is used for predictive maintenance. (3)

[20]

TOTAL: 100

END OF QUESTION PAPER